

Marco Gatti

Ramón y Cajal Fellow
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Current Position

Ramón y Cajal Fellow (tenure track) January 2026–present
Institute of Space Sciences (ICE-CSIC), Universitat Autònoma de Barcelona, Barcelona, Spain.

Previous Positions

KICP Fellow July 2024–December 2025
Kavli Institute for Cosmological Physics, University of Chicago, Chicago, USA.

Postdoctoral Researcher November 2020–July 2024
Department of Physics and Astronomy, University of Pennsylvania, Philadelphia, USA. Advisors: Gary M. Bernstein and Bhuvnesh Jain.

Education

PhD in Physics October 2015–September 2020
Institut de Física d'Altes Energies (IFAE), Universitat Autònoma de Barcelona, Spain. Thesis: *Weak lensing in DES Y3: redshift distributions, shape catalogue, and mass mapping*. Advisors: Ramon Miquel and Chihway Chang. Final mark: Excellent, cum laude.

MSc in Astronomy and Astrophysics October 2012–July 2014
Sapienza University of Rome, Italy. Thesis: *The growth of supermassive black holes: disk instability vs. interactions*. Advisor: Fabrizio Fiore. Final mark: 110/110 cum laude.

BSc in Physics October 2009–July 2012
University of Genoa, Italy. Final mark: 110/110 cum laude.

Selected Talks and Seminars

Simulation-Based Inference for Weak Lensing: From Stage III to Stage IV Surveys, DIPC, Donostia, Spain (**invited**) April 2026

Dimensionality Reduction Techniques for Statistical Inference in Cosmology, University College London (**invited**) December 2024

Cosmology with weak-lensing simulation-based inference and DES, COSMO'24, Kyoto, Japan October 2024

Weak-lensing simulation-based inference, Yale University cosmology seminar (**invited**) March 2024

Non-Gaussian weak-lensing statistics in the era of precision cosmology, Leiden University cosmology seminar (**invited**) December 2023

Non-Gaussian weak-lensing statistics in the era of precision cosmology, Sapienza University of Rome cosmology seminar (**invited**) December 2023

Non-Gaussian weak-lensing statistics in the era of precision cosmology, University of Genoa cosmology seminar (**invited**) December 2023

<i>Cosmology from weak-lensing simulation-based inference</i> , CCA workshop “Debating the potential of machine learning for astronomical surveys”, New York	December 2023
<i>Simulation-based inference with non-Gaussian statistics of weak-lensing observables and DES Y3 data</i> , COSMO23, Madrid	September 2023
<i>Constraints from non-Gaussian weak-lensing analyses</i> , Columbia University cosmology seminar (invited)	March 2023
<i>Cosmology with weak lensing</i> , University of California, Berkeley cosmology seminar (invited)	September 2022
<i>Constraints from non-Gaussian weak-lensing analyses</i> , “Intriguing discrepancies in the growth of structures over cosmic time”, Sexten, Italy (invited)	July 2022
<i>Constraints on AGN feedback models from lensing–tSZ cross-correlations</i> , SZ Workshop, Center for Computational Astrophysics (invited)	June 2022
<i>Cosmology from weak-lensing non-Gaussian statistics</i> , LMU lensing seminar (invited)	May 2022
<i>Cosmology from non-Gaussian map-based statistics with DES Y3 data</i> , IberiCOS, Barcelona	May 2022
<i>Cosmology from non-Gaussian map-based statistics with DES Y3 data</i> , YITP workshop “Cosmology with Weak Lensing: Beyond the Two-point Statistics”, Kyoto (invited)	April 2022
<i>Cosmology from weak-lensing non-Gaussian statistics</i> , Lawrence Berkeley National Laboratory cosmology seminar (invited)	March 2022
Guest expert, workshop “How measuring baryons can improve clustering analyses” (invited)	January 2022
<i>Cosmology from non-Gaussian map-based statistics with DES Y3 data</i> , Cosmopalooza 2022 (invited)	January 2022
<i>Cosmology from non-Gaussian map-based statistics with DES Y3 data</i> , 239th AAS Annual Meeting (invited)	January 2022
<i>Clustering redshifts in DES Y3 and the DES Y3 photo-z calibration strategy</i> , GCCL seminar (invited)	February 2021
<i>Curved-sky weak-lensing mass map with the Dark Energy Survey Y3 data</i> , CMU/UPitt cosmology seminar (invited)	September 2020
<i>Curved-sky weak-lensing mass map with the Dark Energy Survey Y3 data</i> , Duke University cosmology seminar (invited)	January 2020
<i>Cosmological constraints from moments of weak-lensing mass maps</i> , COSMO19, Aachen	September 2019
<i>Cosmology with curved-sky weak-lensing mass maps</i> , KICP cosmology seminar, Chicago (invited)	June 2018
Panelist, photo-z discussion session, “Statistical Challenges for Large-scale Structure in the Era of LSST”, Oxford (invited)	April 2018
<i>Clustering redshifts in the DES Y1 analysis</i> , Euclid photo-z workshop, Toulouse (invited)	October 2017
<i>Triggering AGN activity: disk instability vs. galaxy interactions</i> , MPE seminar, Munich (invited)	April 2015
<i>The role of disk instabilities and galaxy interactions in triggering AGN activity</i> , University of Southampton seminar (invited)	October 2014

Research Visits

Kavli Institute for Cosmological Physics, Chicago, USA — cosmology with moments of weak-lensing mass maps	2018, 2 months
Osservatorio Astronomico di Roma, Rome, Italy — AGN and high-energy astrophysics	2013–2015, 18 months
University of Southampton, UK — HOD and clustering of AGN	2014, 1 week
Institute of Space Astrophysics and Planetology (IAPS), Rome, Italy — star-forming regions in Herschel surveys	2013, 6 months

Scientific Leadership, Membership, and Service

Member, Dark Energy Survey Collaboration	2015–present
Member, Euclid Collaboration	2022–present
Member, NASA OpenUniverse Collaboration	2023–present
Member, South Pole Telescope Collaboration	2025–present
Co-chair, Euclid Forward Modelling Work Package	2024–present
Co-chair, Euclid-US monthly meetings	2024–2025
Co-chair, DES Beyond Two-Point Working Group	2024–present
Co-chair, DES Mass Mapping and Forward Modelling Working Group	2021–2024
Co-chair, DES Shear Catalogue Testing Working Group	2019–2020
Mentor, DES Mentorship Programme	2021–present
External referee for MNRAS, A&A, PASJ, JCAP, ApJ, Open Journal of Astrophysics, and Nature Astronomy	2016–present
External referee, UKRI Funding Service	2024–present
Internal referee, Dark Energy Survey	2016–present

Awards, Fellowships, and Competitive Funding

European Research Council Starting Grant, COSMO-SBI, PI, €1.5M	2026–2031
Plan Nacional project PID2025-172756NB-C31, Co-PI, €273,850	2026–2029
Fondo Italiano per la Scienza (FIS), PI, €1.2M (declined)	2025
Ramón y Cajal tenure-track fellowship, €239,000	2026–2031
KICP Postdoctoral Fellowship, University of Chicago, \$246,000	2024–2027
Postdoctoral Fellowship, University of Pennsylvania \$310,000	2020–2025
GCCL Postdoctoral Fellowship, University of Bochum (declined)	2020
Italian habilitation to academic professorship	2023
Dark Energy Survey Builder status	2021–present
Severo Ochoa “la Caixa” PhD Fellowship	2015–2019
“Wanted the Best” MSc Fellowship, Sapienza University, €5,000	2013
“Percorso di Eccellenza” MSc Fellowship, Sapienza University, €2,000	2013

Computing Allocations

PI, Euclid-US NERSC allocation: 40,000 GPU hours and 5 million CPU hours per year	2025–present
PI, Argonne Leadership Computing Facility: 150,000 GPU hours	2024–present
Co-PI, DiRAC: 140,000 GPU hours	2024–2026

Supervision and Mentoring

Simone Vinciguerra, PhD researcher	2025–present
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Kyle Boone, PhD student, Harvard University	2024–present
Alan Zhou, PhD student, University of Chicago	2024–present
Johnny Pitocco, PhD student, University of Chicago	2024–present
Kunhao Zhong, PhD student, University of Pennsylvania	2023–2024
Dhayaa Anbajagane, PhD student, University of Chicago	2023–2024
Shub Agrawal, PhD student, University of Pennsylvania	2023–2024
Minsu Park, PhD student, University of Pennsylvania	2022–present
Vernon Wetzell, PhD student, University of Pennsylvania	2022–2025
Rafael Gomes, PhD student, University of Pennsylvania	2022–2025
Conghao Zhou, PhD student, University of California, Santa Cruz	2022–2025
Shivam Pandey, PhD student, University of Pennsylvania	2020–2022
Giulia Giannini, PhD student, Universitat Autònoma de Barcelona	2020–2022
Four undergraduate students, University of Pennsylvania	2020–2024

Outreach

Astronomy Conversations Presenter, Adler Planetarium, Chicago — monthly public discussions about astronomy and cosmology	2024–2025
Press release and media interviews for the DES Year 3 analysis	May and October 2021
Hands-on cosmology activities for high-school students visiting IFAE	March 2018 and March 2019
Cosmology seminar for high-school students visiting IFAE	April 2017